MTH-202 Discrete Mathematics

Update MCQS For Mid Term

Solve By Vu Topper RM

85% To 100% Marks

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Range of function \[f(x) = {e^x}\] is................

A. Set of integers
B. Set of negative Real number.
C. Set of positive Real numbers
D. Set of Real number

The logic gate NOT is a uniary operation on {0,1}.

A. false
B. true

If f−1(x)=6−x2 then f−1(2) is ....

A. 2
B. 1
C. 0
D. 3

GIven a Set A={ a,b} and B={1,2,3}, then number of functions from A to B is ....

A. 6
B. 8
C. 5
D. 9

How many one-to-one functions are there from a set with five elements to a set with six elements.

A. 3 factorial (3!)
B. 4 factorial(4!)
C. 6 factorial(6!)
D. 5 factorial(5!)

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

A. true

B. false

Which is not a binary operation on the set of natural numbers N?

A. Multiplication

B. Subtraction

C. None of these.

D. Addition

The relation R = {(12, 14), (13, 5), (-2, 7), (x, 13)} is a function if x is not equal to

A. 14, 7 or 13.

B. 12, 13 or 2.

C. 12, 13 or -2.

D. 14, 5 or 7.

Letf(x)=xandg(x)=−xforallx∈R,then(f+g)(x)is____.

A. -x

B. 0

C. 2x

D. -2x

The Range off:X→Yis also called the image ofXunder

A. true

B. false

Which of the following relations are functions?R1 ={(3, 4),(4, 5),(6, 7),(8, 9)}, R2 ={(3, 4),(4, 5),(6, 7),(3, 9)}, R3 ={(-3, 4),(4, -5),(0, 0),(8, 9)}, R4 ={(8, 11),(34, 5),(6, 17),(8, 19)}

A. R3 and R2 are functions

B. R1 and R3 are functions

C. R1 and R2 are functions

D. R2 and R4 are functions

Cardinality of positive prime numbers less than 20 is ……….

A. 5
B. 10
C. 8
D. 20

Which of the following is not a binary operation on the set of Integers?

A. Multiplication
B. Subtraction
C. Addition
D. Division

A set is countably infinite if,and only if,it has the same cardinality as the set of ....

A. whole numbers
B. real numbers
C. positive integers
D. integers

Let f be a function from X = {2, 4, 5} to Y = {1, 2, 4, 6} defined as f = {(2, 6), (4, 2), (5, 1)}. Then the range of f is _________ .

A. {2, 4, 5}
B. {1, 2, 6}
C. {1, 4, 6}
D. {1, 2, 4, 6}

A function f:R→R defined by f(x)=x−−√ is a real valued function.

A. true
B. false

A graph of a function f is onto iff every horizontal line intersects the graph in at least one point.

A. true

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B. false

Given a set X, define a function i from X to X by i(x) = x from all x belonging to X.Then ____________ .

A. i is both injective and surjective
B. i is injective but not surjective
C. i is surjective but not injective
D. i is neither injective nor surjective

A constant function is one to one iff its …………….. is a singelton.

A. Range
B. Domain

If $f-1(x)=6-x2$ then $f-1(2)$ is ....

A. 2
B. 3
C. 1
D. 0

The logic gates OR and AND are uniary operation on {0,1}.

A. true
B. false

Let $f(x)=x+3$ then $f-1(x)$ is ....

A. $x+3$
B. $x-3$
C. $3-x$
D. $2x-3$

The range of a function is always ....the co-domain of f.

A. a superset of
B. equal to
C. a subset of

D. not equal to

The number of elements in A ×B are ………… if A is a set with ‘5’ elements and B is a set with ‘4’ elements.

A. 0
B. 20
C. 1
D. 9

A function whose range consists of only one element is called……………….

A. One to one function
B. Onto function
C. Constant function
D. Identity function

A set is countable if, and only if, it is _________.

A. infinite
B. finite or countably infinite
C. finite
D. countably infinite

If f(x)=2x and g(x)=x then g(f(x)) is..........

A. 2x
B. 2x2
C. x
D. x2

Let X = {1,2,3,4} and a function ‘f ‘defined from X to X by f(1) = 1 , f(2) = 1, f(3) = 1, f(4) = 1 then which of the following is true?

A. f is an identity function
B. f is injective
C. f is a constant function
D. f is bijective

Domain of the relation {(0,1),(3,22),(90,34)}is ___________ .

A. {1,22,34}
B. {0,1,3}
C. {0,1,3,22,90,34}
D. {0,3,90}

Let P(A) be the power set of a non-empty set A and by using the definition of "subset" relation, ⊆, for all X, Y ∈ P(A), X ⊆ Y ⇔ ∀ x, iff x ∈X then x ∈Y. Then _______.

A. ⊆ is a partial order relation.
B. ⊆ is not a partial order relation.

Let R be the universal relation on a set A then which one of the following statement about R is true?

A. R is not reflexive
B. R is not transitive
C. R is reflexive, symmetric and transitive.
D. R is not symmetric

Let A={1 , 2 , 3 , 4} and R={(1 , 1)(2 , 2)(3 , 3)(4 , 4)} then R is

A. Symmetric
B. Options a,b,and c all true
C. Reflexive
D. Transitive

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is............

A. {(a,1)(2,b)(3,c)(4,d)}
B. {(1,a)(2,b)(3,c)(4,d)}
C. {(a,1)(b,2)(3,c)(4,d)}
D. {(1,a)(b,2)(3,c)(4,d)}

Range of the relation {(0,1),(3,22),(90,34)}is ___________ .

A. {0,3,90}
B. {0,1,3,22,90,34}
C. {1,22,34}
D. {0,1,3}

Let A = {1,2,3} and R be the relation defined on A such that R ={(1,1), (1,2), (2,3), (3,1)}, then R is ________ .
A. Symmetric
B. Antisymmetric

Let R and S be transitive relations on a set A then ______________________ .
A. Both R union S and R intersection S are transitive
B. R intersection S is transitive
C. Neither R union S is transitive nor R intersection S is transitive
D. R union S is transitive

There is atleast one loop in the graph of an irreflexive relation.
A. true
B. false

Let R be a binary relation on a set A.R is anti-symmetric iff _________.
A. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a = b
B. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a ≠ b
C. a, b ∉A if (a,b) ∈R and (b,a) ∈R then a ≠ b
D. None of the above

Find x and y given (2x, x + y) = (6, 2).
A. x=2,y=1
B. x=3,y=-1
C. x=2,y=-1
D. D. x=3,y=1 x=2,y=1

Domain of a relation symbolically written as _______.

A. None of these

B. Dom (R) = {a ∈ A | (a, b) ∈ R}

C. Dom(R) = {b ∈ B | (a, b) ∈ R}

D. Dom (R) = {a ∉ A | (a, b) ∈ R}

Let A = {0,1}and B = {1}.Let R and S be two binary relations on Cartesian product of A and B such that R={(0,1)} and S ={(1,1)}.Then R intersection S = ________________ .

A. {1,1}

B. {(0,1)}

C. {0,1}

D. empty

If x ≡ -10 (mod 15) Which of the following integers are valid solution for x ?

A. 7

B. 5

C. 3

D. 6

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Range of the inverse of the relation is ________.

A. {1, 3, 5, 7}

B. {5, 6, 7, 8}

C. {1, 2, 3, 4}

D. {2, 4, 6, 8}

Let R be a relation on a set A. If R is symmetric then its compliment is ___________ .

A. Irreflexive

B. Symmetric

C. Reflexive
D. Antisymmetric

If A={1 , 2 , 3} is a set and R = {(1 , 2) , (2 , 2) , (2 , 1)} is a relation on A, R is

A. Reflexive
B. Symmetric
C. None of these
D. Transitive

If a relation R={(1,1)(2,1)(1,2)(2,2)} is given then which of the following is not true about this relation.

A. R is Symmetric.
B. R is Transitive
C. R is Reflexive.
D. R is Irreflexive.

If x ≡ 17 (mod 5) Which of the following integers are valid solution for x ?

A. 4
B. -44
C. 8
D. 12

Let R be a relation on a set A. If R is reflexive then its compliment is _____________ .

A. Reflexive
B. Antisymmetric
C. Symmetric
D. Irreflexive

Let the set A = {1,2,3,4} Then the relation {(2,4),(4,2)} is ________.

A. Transitive

B. AntiSymmetric
C. Reflexive
D. Symmetric

The union of the transitive binary relations R = {(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)} and S = {(2,1), (2,4), (3,3)} on the set A = {1,2,3,4} is given by R ∪ S = {(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)}then R ∪ S is_____?

A. not transitive
B. transitive

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Domain of the inverse of the relation is ________.

A. {2, 4, 6, 8}
B. {1, 2, 3, 4}
C. {1, 3, 5, 7}
D. {5, 6, 7, 8}

Let A = {1,2,3,4} and define the relation R on A by R = {(1,2), (2,3), (3,3), (3,4)}.Then ________

A. R is neither reflexive nor irreflexive
B. R is both reflexive and irreflexive
C. R is reflexive
D. R is irreflexive

Range of a relation symbolically written as _______.

A. None of the above
B. Ran (R) = {a ∈ A | (a, b) ∈ R}
C. Ran (R) = {b ∈ B | (a, b) ∈ R}
D. Ran (R) = {b ∉ B | (a, b) ∈ R}

The matrix representation of an inverse relation can be obtained by taking the ....of the relational matrix.

A. **transpose**
B. Determinant
C. inverse
D. adjoint

Let R be a relation on a set A. R is transitive if and only if for all a,b,c∈A then:
A. (a, b) ∈R and (b, c) ∈R but (a, c) ∉R
B. (a, b)∈R then (b, a) ∉R
C. **(a, b) ∈R and (b, c) ∈R then (a, c) ∈R**
D. (a, b) ∈R and (b, c) ∈R then (a, c) ∈R(a, b)∈R then (b, a) ∈R

The properties of being symmetric and being anti-symmetric are _________.
A. Negative of each other
B. **Not negative of each other**

Let A = {1, 2, 3, 4} and define a relation R on A by R = {(1, 1), (1, 2), (1, 3), (2, 3)}.Then which one of the following is a correct statement about R:
A. R is both transitive and reflexive
B. **R is neither reflexive nor transitive**
C. R is transitive
D. R is reflexive

Let A = {0, 1, 2} and R = {(0,2), (1,1), (2,0)} be a relation on A. Then which of the following statement about R is true?
A. R is reflexive
B. **R is symmetric**
C. None of the above
D. R is transitive

Which of the following is always true for the matrix representation of a symmetric relation?

A. Matrix has always 0 in its diagonal entries
B. Matrix has always 1 in its diagonal entries
C. Matrix is singular
D. Matrix is equal to its transpose.

If A=(1,2,3) & B=(4,5,6) and R={(1,4)(2,5)(3,6)(3,4)} The complementary relation is ....

A. A × B
B. A × B (intersection) R
C. A × B (difference or -) R
D. A × B (Union) R

Let A = {1,2,3,4} and R = {(1,2), (2,3), (3,3), (3,4)} be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

A. (3,3)
B. (3,4)
C. (2,3)
D. (1,2)

Range of function /[f(x)=(e^x)/] is __________

- **Set of positive real numbers**

Composite relation symbolically written as ______

- **SoR={(a,c)aeA, ceC, 3eB, (a,b)eR and (b,c)eS}**

If x=17(mod 5) which of the following integers are valid solution for x ?

- **12**

Range of the relation {(0,1),(3,22),(90,34)}

- **{1,22,34}**

Let A= {0,1,2} and R = {(0,2),(1,1),(2,0)} be a relation on A. The which of the following ordered pairs are needed to make it transitive?

- (0,0) and (2,2)

Operation of subtraction is a binary operation on the set of _________

- Integers

Let S=R and define the ‘square’ relation R= {(x,y)Ix^2=y^2}. The square relation is an ________ relation

- Equivalence relation

The logic gate NOT is a uniary operation on {0,1}.

- True

Let A= {1,2}, then P(A)=_________

- {{},{1},{2},{1,2}}

If a relation R is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

- Inverse relation will be irreflexive.

Let R be a binary relation on a set A,R is anti-symmetric iff __________

- a,beA if (a,b)eR and (b,a)eR then a=b

“-“ is a binary operation on the set of integers Z.

- True

The inverse relation R^(-1) from B to A is defined as ___________

- R^(-1) = {(b,a) e B*A I (a,b)eR}

Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix is equal to its transpose

Let A= {1,2,3,4} and let R and S be transitive binary relations on A defined as; R= {(1,2), (1,3), (2,2), (3,3), (4,2), (4,3) and S={(2,1), (2,4), (3,3)} then RuS= {(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)}

- R union S is transitive

Let S=R and define the ' square' relation R = {(x,y)Ix^2=y^2}. The square is an ________ relation.

- Equivalance relation

If x=-10(mod 15). Which of the following integers are valid solution for x?

- 5

Let R be a binary relation on a set A. If R is anti symmetric then ____________

- Inverse of R is anti symmetric

If A={1,2,3} is a set and R = {(1,2),(2,2),(2,1)} is a relation on A, R is

- Symmetric

Let A={0,1} and B=(1). Let R and S be two binary relations on Cartesian product of A and B such that R={(0,1)} and S= {(1,1)}. Then R intersection S=_______________

- Empty

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- Partial order Relation

Let X={1,2,3} and Y={7,8,9} and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- Co-domain of f must contain 1 element

The function fog and gof are always equal

- False

If a relation R={(1,2),(2,3),(3,4)(4,1)(2,2) is given then which of the following is true about this relation.

- R is reflexive

A set is called countable if , and only if, it is___________

- finite

Let $f(x) = x^2-1$ define function f from R to R and c=2 be any scalar, then c,f(x) is _________

- $2x^2+2$

The set Z of all integers is __________

- Countable

Let R be a binary relation on a set A. If R is anti symmetric then ______

- Inverse of R is symmetric

For (2x-3, 4y+2) = (1,10). What will be the value of x and y ?

- (2,2)

Let f and g be the two functions from R to R defined by f(x) = IxI and g(x)= square root of x^2 for all xeR. Then_____

- F(x) is not equal to g(x)

If a set A has 15 elements then P(A) (power set of A) has __________ elements.

- 2^15

For the relation below to be a function, x cannot be what values {(12,14),(13,5),(-2,7),(x,13)}?

- X cannot be 12, 13, or -2

Let the set A = {1,2,3,4}. Then the relation {(2,4),(4,2)} is ___________

- Symmetric

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For the following relation to be a function, x can not be what values? R={(2,4),(x,1), (4,2),(5,6)}.

- x cannot be 2,4 and 5

Vertical line test is used to determine that whether the graph of a relation is a function or not.

- True

The properties of being symmetric and being anti symmetric are ___________

- Not negative of each other

The number of elements in AxB are __________ if A is a set with '5' elements and B is a set with '4' elements.

- 20

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is ____________

- {(1,a)(2,b)(3,c)(4,d)}

Logic gate NOT does not define a binary operation on (0,1) because ___________

- It takes a single input and gives a single output

How many real numbers exist between 1 and 5

- 3

The number pi is

- Irrational

The number square root 2 is

- Irrational

Range the relation {(0,1)(3,22),(90,34)}

- {0,3,90}

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Supherical coordinate 0 is related to the cylindrical coordinate as____________

- @=@

Operation of subtraction is a binary operation on the set_______

- Integers

Let A {1,2,3,4} and R={(1,2),(2,3),(3,3),(3,4)} be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

- (3,3)

Input values of the function are called the ___________

- Domain

Range of function f(x)=IxI will be

- Set of positive real numbers

Which of the following is not a binary operation on the set of integers?

- Division

In the matrix representation of an irreflexive relation all the entries in the main diagonal are _________

- 0

If the partition set of A is {A_1,A_2} then

- A_1nA_2= not empty set

Let A= {a,b} then P(A) =

- {Non empty set, {a},{b},{a,b}}

Which relations below are not functions?

- {(13,14),(13,5),(16,7),(18,13)}

In the directed graph of an antisymmetric relation there is _______ pair of arrows between two distinct elements of the set.

- No

If a relation R= (1,1),(2,1)(2,2) is given then which of the following is not true about this relation

- R is irreflexive

Let R and S be transitive relations on a set A then _____________

- Neither R union S is transitive nor R intersection S is transitive

Let R={(1,2)(3,4)(5,6)(7,8)}. Domain of the inverse of the relation is __________

- {2,4,6,8}

Let A={1,2,3,4,5} and B={4,9,,16,17,25}. Then the relation R={(2,4),(3,9),(4,16),93,17)} The inverse of R is'

- {(4,2),(9,3),(16,4),(17,3)}

Let R be a relation on a set A. If R is symmetric then its compliment is ________

- Irreflexive

Which is not a binary operation on the set of natural numbers N?

- Subtraction

If a relation R={(1,1)(2,1)(1,2)(2,2)} is given then which of the following is not true about this relation.

- R is irreflexive

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is ___________

- {(1,a)(2,b)(3,c)(4,d)}

For any set A, the Cartesian product of A and A is known as ___________

- Universal relation

Let A={p,q,r,s} and define a relation R on A by R={(p,p),(p,r),(q,r),(q,s),(r,s)} Then which one of the following is the correct statement about R:

- R is not reflexive

A={1,2} B={3,4}, R={(1,3)(2,4)}. Then the complement of R is _________.

- {(1,4)(2,3)}

Domain of a relation symbolically written as___________.

- Dom(R)={aeAI(a,b)eR}

Let X={2,4,5} and Y={1,2,4} and R be a relation from X to Y defined by R={(2,4)(4,1)(a,2)}. For what value of 'a' the relation R is a function?

- 5

Let A={1,2,3,4,5,6,7,8,9}, then which of the following sets represent the partition of the set A?

- A={1,3,5,7,9}, B={2,4,6}, C={8}

Let A={1,2,3} and B={2,4} then number of binary relations from A to B are ____________.

- 64

A relation R is said to be _______iff it is reflexive, antisymmetric and transitivde.

- Partial order Relation

Let f be a function from X={2,4,5} to Y={1,2,4,6} defined as:f={(2,6),(4,2),(5,1)} . The range of f is ________

- {1,2,6}

Let A={0,1,2} and R={(0,2),(1,1),(2,0)} be a relation on A. Then which of the following statement about R is true?

- R is symmetric

Let A={2,3,4} and B={2,6,8} and let R be the "divides" relation from A to B i.e for all (a,b) belong to (Cartesian product of A and B), a,R b iff a I b (a divides b). Then

- R={(2,2),(2,6),(2,8),(3,6),(4,8)}

Let A={1,2,3,...,50} and B={2,4,6,8,10}. Then the Cartesian product of A and B has __________ elements.

- 250

In the matrix representation of an reflexive relation all the entries in the main diagonal are __________

- 1

Which of the following is not a type of a relation?

- Permutation

Let X={2,4,5} and Y={1,2,4} and R be a relation from X to Y defined by R={(2,4),(4,1),(a,2)}. For what value of 'a' the relation R is a function ?

- 5

Which of the following is not a representation of a relation?

- Venn diagram

Let A={1,2,3,4} and define the following relations on A. Then R={(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)} is ________

- R is irreflexive

The range of f:X-> Y is also called the image of

- True

Complementary Relation symbolically written as ________

- R=A*B- R={(a,b)eA*BI (a,b)not belong to R}

Let A={1,2,3,4} and R=(1,1)(2,2),(3,3),(4,4) then R is

- All options

If a relation R={(1,1),(2,1),(1,2),(2,2)} is given then which of the following is not true about this relation.

- R is irreflexive

Which of the following logical connective is not a binary operation ?

- Implication

A set may be dividend up into its disjoint subsets, such division is called______

- Partition

If A=(1,2,3)&B=(4,5,6) and R ={(1,4)(2,5)(3,6)(3,4)} The complementary relation is ________

- A*B(difference or –) R

In matrix representation of a_________ relation, the diagonal entries are always 1.

- Reflexive

R is not symmetric iff there are elements a and b in A such that __________

- (a,b) belongs to R but (b,a) does not belong to R

Which relations below are functions?

R1={(3,4),(4,5),(6,7),(8,9)}

R2={(3,4),(4,5),(6,7),(3,9)}

R3={(-3,4),(4,-5),(0,0),(8,9)}

R4={(8,11),(34,5),(6,17),(8,19)}

- R1 and R3 are functions

The logic gate OR and AND are uniary operation on {0,1}

- False

There is atleast one loop in the graph of an irreflexive relation

- False

There is atleast one loop in the graph of an reflexive relation

- True

A contains 3 elements and B contains 2 elements, then number of subsets of A*B are ________

- 64

Let A={1,2,3} and B={0,1,2} and C={a,b} R={(1,0),(1,2),(3,1),(3,2)} S={(0,b),(1,a),(2,b)} composite of R and S=____________

- {(1,b),(1,a),(3,a),(3,b)}

If R is transitive then the inverse relation will be transitive

- True

The number of elements in the power set of P(not empty set) denoted by P(P(not empty set) is

- 2

The function defined from Z to Z as f(x)= 1/(x+2)(x-2) is not well defined because _________

- Function is not defined at x=2 and x=-2

Rang of relation {(0,1),(3,22),(90,34)} is _________

- {1, 22, 34}

The number of elements in the power set of P(not empty set) denoted by P(not empty set) is

- 1

Let R={(1,2),(3,4),(5,6),(7,8)}. Domain of inverse of the relation is ________

- {2,4,6,8}

The relation 'divides' on the set of integers is ________

- A symmetric relation

Operation of subtraction is a binary operation on the set of_________

- Integers

If R is transitive then the inverse relation will be transitive.

- True

Let A={1,2,3,4} and define the relation R on A by R={(1,2),(2,3),(3,3),(3,4)}. Then_________

- R is both reflexive and irreflexive

A set may be divided up into its disjoints subsets,such division is called

- Partition

If a set A contains is elements then the number of elements in its power set P(A) is _______

- 2^n

Range of a relation symbolically written as ______.

- Ran R = {b ∈ B I (a, b) ∈ R }

Let R be a binary relation on a set A. IF R is anti symmetric then ________.

- Inverse of R is anti symmetric

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Let A be a set with m elements and B be a set with n elements then the number of elements in A*B are ________

- m.n

Let A{1,2,3,4} and define the following relation on A. Then R={(1,3),(2,2),(2,4),(3,1),(4,2)} Is _________

- R is symmetric

If A={1,2,3}& B={4,5,6} and R={1,4},{2,5},{3,6},{3,4}. The complementary relation is _______

- A*B (difference or -) R

Let R be a relation on a set A. If R is reflexive then its compliment is _________

- Irreflexive

25=1(mod 3) means that 3 divides ______

- 25-1

Let R be the universal relation on a set A then which one of the following statement about R is true ?

- R is reflexive, symmetric and transitive

Domain of a relation symbolically written as _____

- Dom(R)= {aeR I (a,b) e R}

Let R be a relation on a set A. R is transitive if and only if for all a,b,ceA then

- (a,b)eR and (b,c)eR then (a,c)eR

Let A be a non-empty set and P(A) the power set of A Deifne the ' subset' relation ,c as follows for all X,Y e P(A), XcY <-> for all x, iff xeX then xeY . Then c is ________

- c is partial order relation

Define a relation R={(1,1),(2,2),(3,3),(1,3) the relation is

- R is reflexive and transitive

Let R be a relation on a set A. R is transitive if and only if for all a,b,ceA then.

- (a,b)eR and (b,c)eR then (a,c)eR

Let R and S be reflexive relations on a set A then R intersection S is reflexive

- True

A function whose range consists of only one element is called_______

- One to one function

The set Z of all integers is _________

- Countable

One-to-one correspondence means the condition of ______

- Both (a) and (c)

Let X={1,5,9} and Y={3,4,7}, Define a function f from X to Y such that f(1)=7,f(5)=3,f(9)=________. Which is true f(9) to make it a one-to-one (injective) function?

- 4

What will be the fourth term of the following sequence 1/2,2/3,3/4 ______?

- 4/5

The value of 6!=

- 720

A constant function is one to one iff its _______ is a singleton.

- Domain

A constant function is onto iff its _________ is a singleton.

- Co-domain

Number of one to one functions from X={a,b} to Y={u,v} are equal to _____-

- 2

If f is defined recursively by f(0) = -1 and f(n+1)=f(n)+3, then f(2)=________.

- 5

A function whose inverse function exists is called a/an_________

- Invertible function

Let f(2)=3, g(2)=3, f(4)=1 and g(4)=2 then the value of fog(4) is......

- 3

Let A={1,2,3,4} and B={7} then the constant function from A to B is _______.

- Onto

Composition of a function is a commutative operation.

- False

Composition of a function is not a commutative operation

- True

The sum of first five whole number is _________.

- 10

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- True

Inverse of a surjective function is always a function.

- False

Inverse of a surjective function may not be a function

- True

Let X={1,2,3,4} and Y={7,8,9} and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- Co-domain of f must contain 3 elements

If f; X->Y and g; Y->Z are both onto functions. Then gof; X->Z is ________

- onto

If f and g are two one-to-one functions, then their composition gof is________

- One-to-one

If f: W →X, g:X →Y, and h:Y →Z are functions, then__________

- (hog)of = ho(gof)

Cardinality of positive prime numbers less than 20 is _________.

- 8

IF f(x)=sin^{-1}(x) and g(x)=sin x then gof(x) is ________.

- X

0!=________.

- 1

An important data type in computer programming consists of _________.

- Finite sequences

Let f(x)=2x and g(x)=x+2 define functions f and g from R to R, then (f-g)(x) is __________.

- x-2

The total number of terms in an arithmetic series 0+5+10+15+….50 are ________.

- 11

9!/6!=_______

- 504

Let f(x)=3x and g(x)=3x-2 define functions f and g from R to R, THEN (F+G)(X) is __________

- 6x-2

If f is a bijective function then $(f^{-1}f(x))$ is equal to

- X

A sequence whose terms alternate in sign is called an________

- Alternating sequence

Common ration in the sequence "4, 16, 64, 256,…" is…..12.

- 4

0.8181818181 is a infinite geometric series.

- True

The word 'algorithm' refers to a step-by-step method for performing some action.

- True

A predicate become _____ when its variables are given specific values

- Sentence

The sum of two irrational numbers must be an irrational number.

- False

The sum of two irrational numbers need not be irrational number

- True

The division by zero is allowed in mathematics.

- Fasle

The product of any two consecutive positive integers is divisible by 2

- True

If ‘n’ is an odd integer then n^3+n is _______.

- Even

For integers a,b,c, If divides b and a divides c, then a divides (a+b).

- False

Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that _______ and $0<\underline{r}<d$

- N=d.q+r

A rule that assigns a numerical value to each outcome in a simple space is called

- Random variable

If A and B are two disjoint (mutually exclusive) events then P(AB)=

- P(A) + P(B)

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

- C(10,5)

The expectation of x is equal to

- Sum xf(x)

If P(A intersection B) = P(A) P(B) THEN THE events A and B are called

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- Independent

A walk that starts and ends at the same vertex is called.

- None optins

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

- 497

What is the probability of getting a number greater than 4 when a die is thrown?

- 3/5

Eater formula for graphs is__________.

- F=e-v+2

X+a,x+3a,x+5a.....is an___________

- Arithmetic sequence

Composition of a function is a commutative operation.

- False

Real valued function is a function that assigns _____ to each member of its domain.

- Only a real number

Let X={1,2,3,4} and a function 'f' defined on X f(1)=1,f(2)=2,f(3)=3,f(4)=4 then ________

- F is an identity function

A constant function is surjective if and only if ___________

- The co-domain consists of a single element

Cardinality means the total number of elements in a set.

- True

If $f(x)=2x$ and $g(x)=x$ then $g(f(x))$ is ________.

- **$2x^2$**

Let f:R->R is one to one function then c,f, c is not equal 0 is also one to one function.

- **True**

Let X={1,2,3,4} and a function 'f' defined from X to X by f(1)=1, f(2)=1, f(3)=1, f(4)=1 then which of the following is true?

- **F is a constant function**

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- **True**

Which of the following is not correct for a 'sequence'?

- **A sequence is a relation whose domain is the set of natural numbers**

$F(x)=x^2$ is not one to one function from R to R^+

- True

Let f: R->R is one to one function then c,f c is not equal to is also one to one function.

- **True**

Let $f(x)=x+2$ then $f^{-1}(x)$ is_______

- **x-2**

Let $f(x)=x^2+1$ define functions f from R to R and c=2 be any scalar, then c,f(x) is _________.

- **$2x^2-1$**

One to one correspondence means the condition of ________.

- Both (a)and (c)

A function F: R-> R defined by f(x) = square root x is a real valued function.

- False

If g:R->R defined by $g(x)=e^2$ is a real valued function of a real variable.

- True

A function F:R-> R defined by f(x) = log x is a real valued function

- True

½, then 3rd term of sequence is ________.

- 1/2

The process of defining an object in terms of smaller versions of itself is called recursion.

- True

Which of the following is not correct for a 'sequence'?

- A sequence is a relation whose domain is the set of natural numbers

A set is called countable if, and only if, it is __________.

- Finite and countable infinite.....both

A set that is not countable is called _________.

- Uncountable

A sequence whose terms alternate in sign is called an ______.

- Alternating sequence

Let f: R->R is one to one function then c,f is also one to one function for ______.

- C is not equal 0

Let f(x) = x+3 then $f^{-1}(x)$ is _________-

- x-3

Let f and g be two functions defined by f(x) = x+2 and g(x)= 2x+1. Then the composition of f and g is_________.

- 2x+3

Number of one to one functions form X={a,b} to Y={u,v} are equal to _____

- 2

The flbonacci sequence is deined as $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$ for all integers $k \geq 2$ then which of the following is true for F_2

- F_2-F_1= 2+1=3

x+a, x+3a, x+5a,...... is a/an_____.

- Arithmetic sequence

Inverse of a function may not be a function.

- True

In the following sequence a_k=K/(k+1), for k=1, a_1 will be _________.

- ½

If f(x)=x and g(x)= -x are both one to one function then (f+g)(x) is also one to one function.

- False

If f(x)=x and g(x)= -x are both one to one function then (f+g)(x) is not one to one function.

- True

The function ‘f’ and ‘g’ are inverse of each other if and only if their composition gives______.

- Identity function

Which of the following set is the domain of a sequence?

- Set of real numbers

Let C is defined as the set of all countries in the world then C is a ________.

- Finite set

A constant function is surjective if and only if_______.

- The co domain consists of a single element

The sum of the series $a_1 + a_2 + a_3 + \ldots\ldots$ can be written as __________.

-

Inverse of a function may not be a function.

- True
- 3

Let X={1,2,3,4} and a function ‘f’ defined from X to X by f(1)=1, f(2)=1, f(3)=1, f(4)=1 then which of the following is true?

- F is a constant function

The composition of function is always

- Associative

A set is countably infinite if, and only if, it has the same cardinality as the set of

- Positive integers

Two functions ‘f’ and ‘g’ from ‘X’ to ‘Y’ are said to be equal if and only if _____.

- F(x)=g(x) for all ‘x’ belongs to X

Y=x^3 is a graph of bijective function from R to R.

- True

Domain and range are same for__________.

- Identity funtion

Let X={1,2,3,4} and a function ‘f’ defined on X by f(1)=1,f(2)=2,f(3)=3

- F is an identity function

Composition of a function is a commutative operation.

- True

Inverse of a surjective function is always a function.

- False

A function whose inverse function exists is called a/an_______.

- Invertible

Given a set X define a function I from X to X by i(x)=x from all x belonging to X . Then________.

- I is both injective and surjective

Let f: R->R is a one to one function then c,f is also one to one function for.

- C is not equal 0

Let X={1,5,9} and Y={3,4,7}. Define a function f from X to Y such that f(1)=7, f(5)=3, f(9)=_____. Which is true for f(9) to make it a one-to-one (injective) function?

- 4

Which of the following is not a predecessors of ak?

- A_k+1

Two functions ‘f’ and ‘g’ from X to Y are said to be equal if and only if________.

- F(x)=g(x) for all ‘x’ belongs to x

The two functions ‘f’ and ‘g’ are equal if ______.

- F(x) =3x and g(x)= $6x^2+3x/2x^2+1$ for all xeR

If first term of a geometric sequence is 2 and common ratio is ½, then 3^{rd} ter, of sequence.

- ¼

Y= squre root x is an _________function form R^+ to R

- One to one function

Y= x^2 is an _________function form R to R^+

- NOT ONE TO ONE FUNTION

If a function (gof)(x) : X->Z is defined as (gof)(x)=g(f(x)) for all xeX, Then the function_________—.

- (gof)

If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is ________.

- -20

If f(x)=$\sin^{-1}$(x) and g(x) = sin x then gof(x) is __________.

- X

F: X->Y that is both one to one and onto is called a ________.

- Bijective function

What does ‘y’ denotes in a geometric sequence?

- Common ratio

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Let g be a function defined by g(x)=x+1. Then the composition of (gog).

- X+2

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Which of the following is true for the following sequence?

- If n is even, then Cn=2 and if n is odd, then Cn = 0

The function 'f' and 'g' are inverse of each other if and only if their composition gives__________.

- Identity function

N! is defined to be __________.

- The product of the integers from 1 to n

Let A = {1,2,3,4} and B={7} then the constant function from A to B

- Both one to one and onto

A set is called countable if, and only if, it is __________.

- Countably infinite and finite

If f(x) = x and g(x) = -x are both one to one functions then (f+g)(x) is also one to one function.

- False

If a is the 1st term and d be the common difference of an arithmetic sequence then the sequence is a, a+d, a+2d, a+3d....

- True

x+a, x+3a, x+5...... is a/an________.

- Arithmetic sequence

inverse of an injective function may not be a function.

- True

$y=x^3$ is a graph of bijective function form R to R

- False

Two functions 'f' and 'g' from x to y are said to be equal if and only if _____.

- F(x)=g(x) for all 'x' belongs to x

Common ration in sequence '36, 12, 4 , 4/3, ...' is ......

- 1/3

A set is called finite if, and only if, is the ______ or there is ———.

- Empty set or one-to-one

Let $f(X)=x^2-1$ and g(x)=x+1 define functions f and g from R to R, then (f/g)x

- x-1

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Let g be a function defined by g(x) = x+1. Then the composition of (gog).

- X+2

F:X->Y that is both one to one and onto is called a __________.

- Bijective function

What does 'I' denotes in a geometric sequence?

- Common ratio

Let A={1,2,3,4} and B={7} then the constant function from A to B is _____.

- Both one to one and onto

N! is defined to be _______.

- The product of the integers from 1 to n

A set is called countable if, and only if, it is ________.

- Both b and c

Which of the following is the example of an alternating sequence?

- C_n=n/n+1 for n $\geq$ 0

X+a, x+3a, x+5a….. is an_______

- Arithmetic sequence

0, -5, -10, -15, … is an _________.

- Arithemetic sequence

5, 9, 13, 17, … is an _________.

- Arithemetic sequence

An important data type in computer programming consists of ___________.

- Finite sequence

One-to-one correspondence means the condition of ________.

- One-one and onto ….both

Cardinality means the total number of elements in a set.

- True

Inverse of a function may not be a function.

- True

The functions ‘f’ and ‘g’ are inverse of each other if and only if their composition gives ___________.

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- Identity function

A set is called finite if , and only if, it is the ________ or there is _______.

- Empty set or one-to-one

Let f and g be the two functions from R to R defined by f(x) = IXI and g(x) = square root x^2 for all xeR, then _________.

- F(x) is not equal to g(x)

If $f^{-1}(x)$ = 6-x/2 then f^{-1} (2) is _________.

- 2

Let f(2)=3, g(2)=3, f(4)=1 and g(4)=2 then the value of fog(4) is ______.

- 3

An important date type in computer programming consists of _——.

- Finite sequence

Let f(x)=x+3 then f^{-1}(x) is ________.

- X-3

If f:X->Y amd g:Y->Z are both onto function. Then gof : X->Z is ______.

- One-to-one function

The functions 'f' and 'g' are inverse of each other if and only if their composition gives

- Identity function

The two function 'f' and 'g' are equal if ___________.

- F(x) = 3x and g(x) = $6x^2$+3x/2x+1 for all xeR

Two functions 'f' and 'g' from X To Y are said to be equal if and only if——.

- F(x) and g(x) for all 'x' belongs to X

Which of the following set is the domain of a sequence?

- Set of natural numbers

If 1st term of a geometric sequence is 2 and common ratio is 1/2 , then 3rd term of sequence is _________.

- 1/2

Composition of a function is a commutative operation.

- True

If a function (gof)(x): X-> Z is defined as (gof)(x)=g(f(x)) for all xeX. Then the function ________ is known as composition of f and g.

- (g o f)

A set is countably infinite if and only if and only if, it has the same cardinality as the set of ________.

- Positive integers

The 3rd term of the sequence $b_n=5^n$ is ______.

- 125

If f is a bijective function then (f^{-1}(f(x)) Is equal to _____.

- X

Let f(x) = x and g(x) = -x for all xeR, then (f+g)(x) is ____.

0

An infinite sequence may have only a finite number of values.

- True

The functions fog and gof are always equal .

- False

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- True

A function whose range consists of only one element is called __________.

- Constant function

Let X ={1,5,9} and Y={3,4,7}. Define a function f from X to Y such that f(1)=7, f(5)=3, f(9)=4 then which of the following statement about 'f' is true?

- F is both one-to-one and onto

$Y=x^3$ is a graph of bijective function from R to R.

- True

A function whose inverse function exists is called a/an_____.

- Invertible

Let F and g be two functions defined by f(x)= x+2 and g(x)= 2x+1. Then the composition of f and g is _____.

- 2x + 5

If r is a positive real number, then the value of r in 3, r,r=-27r is

- -9

The _____ of the terms of a sequence forms a series.

- Sum

The sum of first five whole number is ________.

- 10

If f_k=f_k-1+f_k-2then f_0=1 , f_1=2, then f_2= _________.

- 3

Let f(x) = 3x and g(x) = x + 2 define functions f and g from R to R, then (f.g)(x) is _____.

- $3x^2 + 6x$

Let R be a relation on a set A. If R is reflexive then its compliment is _______ .

- Irreflexive

If A = Set of students of virtual university then A has been written in the ________.

- Descriptive form

If a function (g o f)(x):X→Z is defined as (g o f)(x) = g(f(x)) for all x ∈ X. Then the function _______ is known as composition of f and g.

- (g o f)

If X and Y are independent random variables and a and b are constants, then Var(aX + bY)is equal to

- aVar(X) + bVar(Y)

Let A = {1, 2, 3} and B = {2, 4} then number of functions from A to B are ________.

- 8

p is equivalent to q' means _______.

- p is necessary and sufficient for q.

Let A and B be subsets of U with n(A) = 12, n(B) = 15, n(A') = 17, and n(A intersection B) = 8, then n(U)=_____ .

- 29

For the following relation to be a function, x can not be what values?
R = {(2,4), (x,1), (4,2), (5,6)}

- x cannot be 2, 4 or 5

Find the number of the word that can be formed of the letters of the word “ELEVEN”.

- 120

There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

- 6

Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

- 16

A predicate becomes ________ when its variables are given specific values.

- statement

Find the number of distinct permutations that can be formed using the letters of the word ”BENZENE”

- 420

Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- 40

In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

- 24

If p ↔ q is True, then _______.

- p and q both are True.

If A and B be events with $P(A) = 1/3$, $P(B) = 1/4$ and $P(A \cap B) = 1/6$, then $P(A \cup B) =$ _______ .

- 5/12

an integer n is a perfect square if and only if _______ for some integer k.

- n = k^2

If A and B are disjoint finite sets then $n(A \cup B) =$ _____.

- n(A) + n(B)

Let X = {2, 4, 5} and Y= {1, 2, 4} and R be a relation from X to Y defined by R = {(2,4), (4,1), (a,2)}. For what value of ‘a‘ the relation R is a function ?

- 5

$\sim(P \rightarrow q)$ is logically equivalent to ________.

- $p \wedge \sim q$

A tree is normally constructed from _______.

- left to right

A Random variable is also called a ________.

- Chance Variable

The conjunction $p \wedge q$ is True when ________.

- p is True, q is True

The logical statement $p \wedge q$ means _______.

- p AND q

Which of the followings is the factorial form of 5 . 4?

- 5!/3!

What is the minimum number of students in a class to be sure that two of them are born in the same month?

- 13

If p is false and q is true, then $\sim p \leftrightarrow q$ is _______.

- True

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- TRUE

($p \vee \sim p$) is the _______.

- Tautology

(-2)! = ________ ?

- Undefined

If p = It is raining, q = She will go to college
"It is raining and she will not go to college"
will be denoted by

- $p \wedge \sim q$

Let X = {1, 2, 3}, then 2-combinations of the 3 elements of the set X are ________?

- {1, 2}, {1, 3} and {2, 3}

Let $f(x) = x^2 + 1$ define functions f from R to R and c = 2 be any scalar, then c.f(x) is _____.

- $2x^2 + 2$

The disjunction of p and q is written as _______.

- $p \vee q$

If X and Y are independent random variables, then E(XY) is equal to

- E(x)E(y)

How many possible outcomes are there when a fair coin is tossed four times?

- 16

Which of the followings is the product set A * B * C ? where A = {a}, B = {b}, and C = {c, d}.

- {(a, b, c), (a, b, d)}

The number of the words that can be formed from the letters of the word,"COMMITTEE" are

- 9! / (2!2!2!)

One-to-One correspondence means the condition of ______.

- One-One and onto

The functions f o g and g o f are always equal.

- FALSE

If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- K-Sample

Determine values of x and y, where (2x, x + y) = (8, 6).

- x = 4 and y = 2

Let g be a function defined by g(x) = x + 1. Then the composition of (g o g)(x)is ______.

- x + 2

What is the truth value of the sentence?
'It rains if and only if there are clouds.'

- False

Reductio and absurdum' is another name of ________.

- Proof by contradiction

X belongs to A or x belongs to B, therefore x belongs to _______.

- A union B

Which of the followings is the product set A * B * C? Where A = {a}, B = {b}, and C = {c, d}.

- {(a, b, c), (a, b, d)}

Real valued function is a function that assigns ______ to each member of its domain.

- Only a real number

The negation of "Today is Friday" is

- Today is not Friday

A non-zero integer d divides an integer n if and only if there exists an integer k such that ________.

- n = d k

The statement p → q is logically equivalent to ~q → ~p

- True

Let R be the universal relation on a set A then which one of the following statement about R is true?

- R is reflexive, symmetric and transitive.

Let f(x)=3x and g(x) = 3x − 2 define functions f and g from R to R. Then (f+g)(x) = _______.

- 6x − 2

The switches in parallel act just like _______.

- OR gate

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The converse of the conditional statement p → q is

- q → p

If X and Y are random variables, then E(aX) is equal to

- aE(X)

Which of the following statements is true according to the Division Algorithm?

- 17 = 5 x 3 + 2

Let p → q be a conditional statement, then the statement q → p is called _______.

- Converse

The disjunction p ∨ q is False when _______.

- P is False, q is False.

A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

- 30

The converse of the conditional statement 'If I live in Quetta, then I live in Pakistan' is _______.

- If I live in Pakistan, then I live in Quetta.

The functions 'f' and 'g' are inverse of each other if and only if their composition gives ______.

- Identity function

P (0, 0)=_____?

- 1

Let p1, p2, p3 be True premises in a given Truth Table. If the conjunctions of the Conclusion with each of p1, p2, p3 are True, then the argument is _______.

- Valid

If p is false and q is false, then ~p implies q is _______.

- False

A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement?

- 125

Let A = {2, 3, 5, 7}, B = {2, 3, 5, 7, 2}, C = Set of first five prime numbers. Then from the following which statement is true ?

- A = B

The set of prime numbers is ________.

- Infinite set

The contrapositive of the conditional statement 'If it is Sunday, then I go for shopping' is _______.

- I do Not go for shopping, then it is Not Sunday.

Let p be True and q be True, then (~p ∧ q) is _______.

- False

In how many ways a student can choose a course from 2 science courses,3 literature courses and 5 art courses.

- 30

The method of loop invariants is used to prove _________ of a loop with respect to certain pre and post-conditions.

- correctness

A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions

- **126**

If A and B are any two sets, then A − B = B – A

- **False**

There are 5 girls students and 20 boys students in a class. How many students are there in total?

- **25**

A sequence whose terms alternate in sign is called
Series

Cauchy sequence

Alternating sequence

None of the above

The 3rd term of the sequenceB5=5n

125

625

5

25

If a function is recursively defined as f(0) = 4 & f(n + 1) = 3f (n) + 3 then f(2)=..........

15

55

48

50

The _____ of the terms of a sequence forms a series.

Sum

Multiplication

Let f:R->R is one to one function then c,f, c is not equal 0 is also one to one function.

False

True

720 = ...

4!

5!

7!

6!

The composition of functions is always _____.

Onto

Associative

one-to-one

commutative

What will be the fourth term of the following sequence 1/2 , 2/3 , 3/4 , ___?

5/6

7/8

4/5

6/7

Which of the following set is the domain of a sequence?

Set of natural numbers

Set of complex numbers

Set of real numbers

Set of rational numbers

Let f(X)=x2-1 and g(x)=x+1 define functions f and g from R to R, then (f/g)x

X2-1

X2+1

x-1

x+1

9!/6!=_____.

506

509

504

509

If a function is recursively defined as f(0) = 4& F(n + 1) = 3f (n) + 3 then f(1) =____

25

10

20

15

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